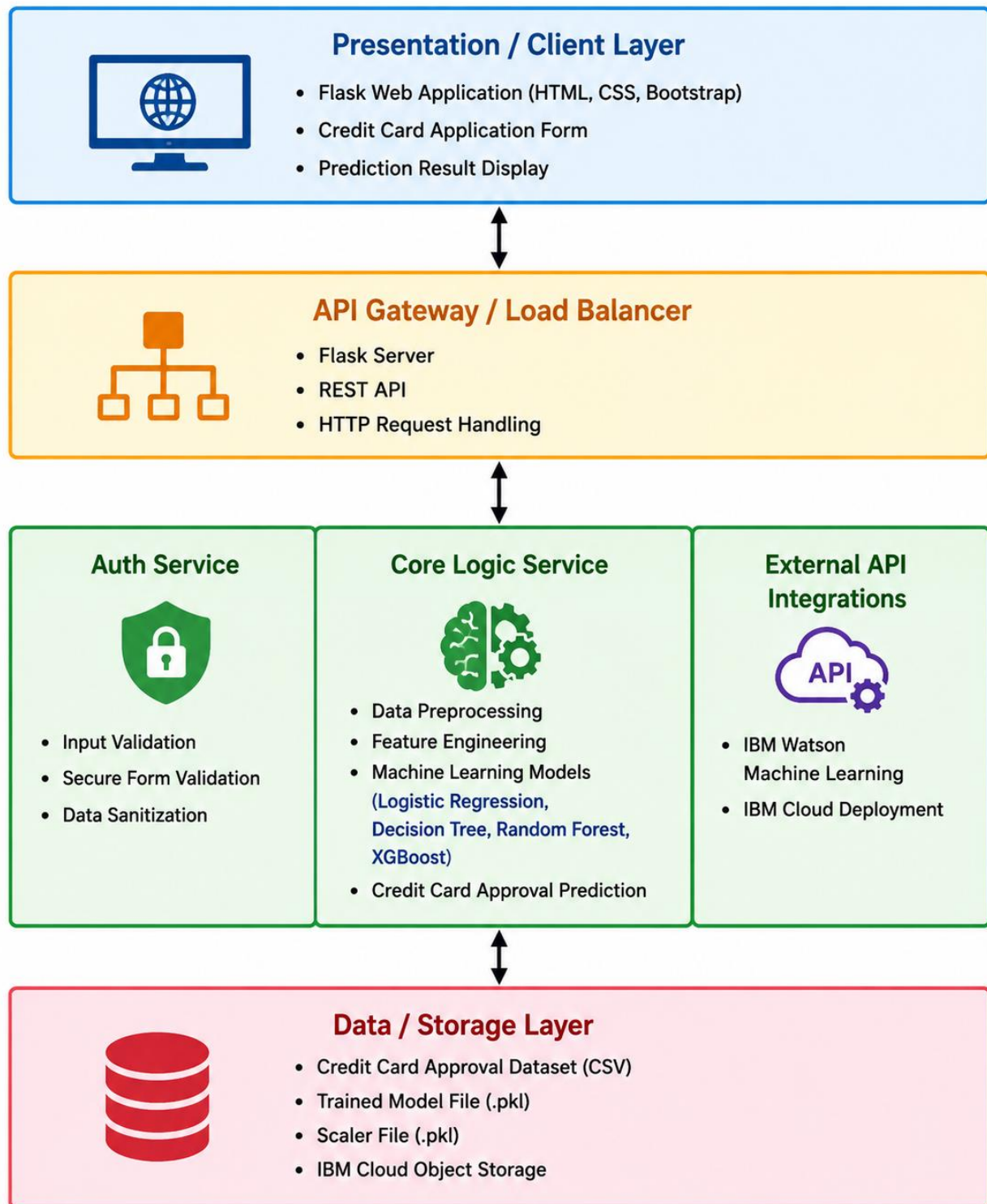


Solution Architecture

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

The architecture of the **Credit Card Approval Prediction** system is organized into three layers: the **Presentation Layer**, **Application Layer**, and **Data Layer**. The Presentation Layer provides a user-friendly Flask web interface where users enter applicant details and receive instant approval predictions. The Application Layer processes user requests, performs data preprocessing, executes machine learning models (Logistic Regression, Decision Tree, Random Forest, and XGBoost), and generates the prediction result. The Data Layer stores the credit card dataset, trained machine learning model, scaler, and related files. The solution also integrates with **IBM Watson Machine Learning** for cloud deployment, enabling secure, scalable, and real-time credit card approval predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides a web interface for users to enter applicant details and displays the approval/rejection result.	Flask, HTML, CSS, Bootstrap
<i>API Gateway</i>	<i>Receives user requests, validates inputs, and sends data to the prediction service.</i>	Flask, REST API, HTTP
Core Logic Service	<i>Performs data preprocessing, feature engineering, loads the trained ML model, and predicts credit card approval.</i>	Python, Scikit-learn, XGBoost, Pandas, NumPy
Database / Storage	Stores the credit card dataset, trained model (.pkl), scaler, and application files.	CSV Dataset, Joblib/Pickle, IBM Cloud Storage